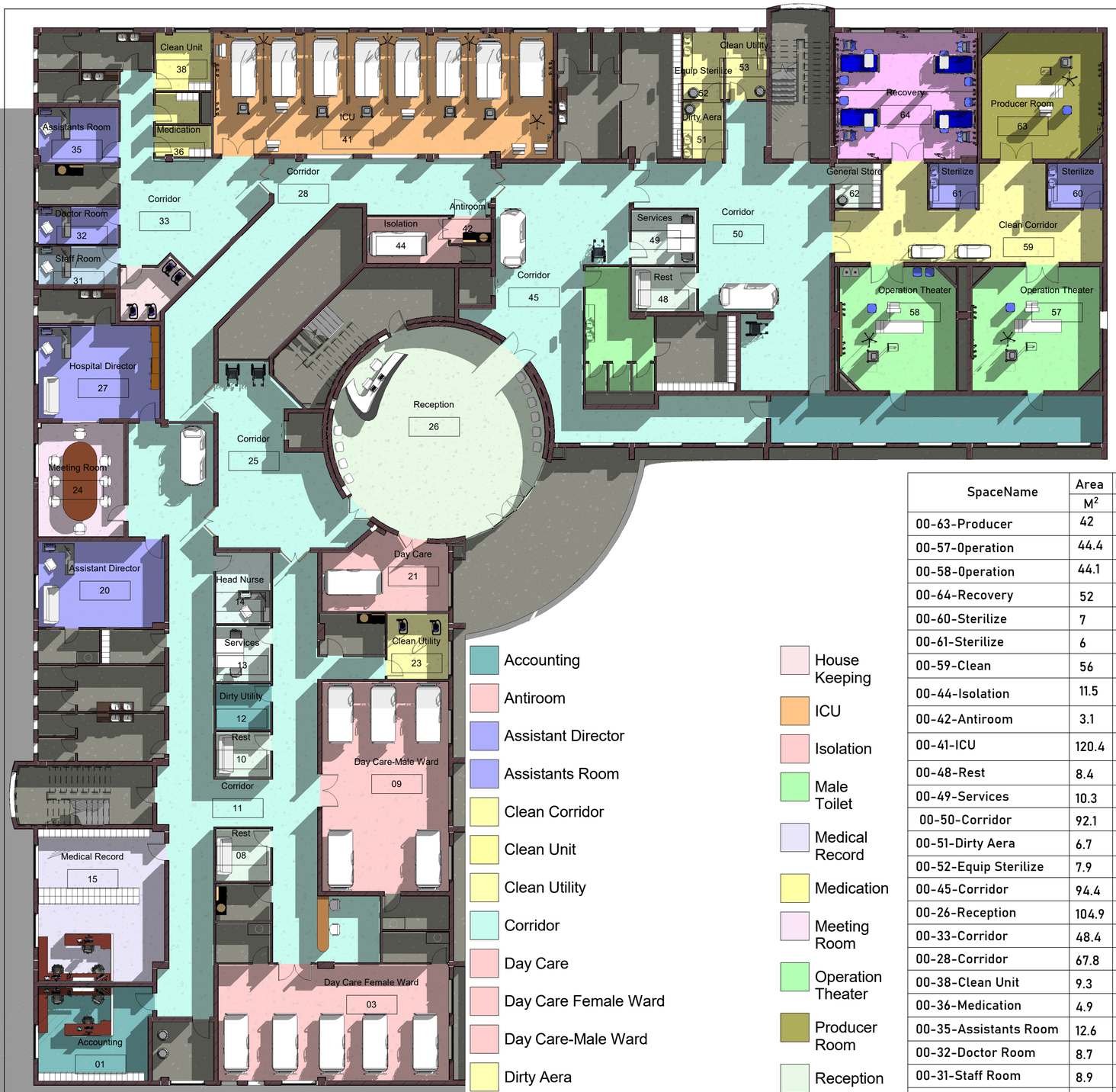


1 Ground Floor Load HAP

1 : 170

CHILLER SELECTION:

Total Building Cooling Load = 108 TR

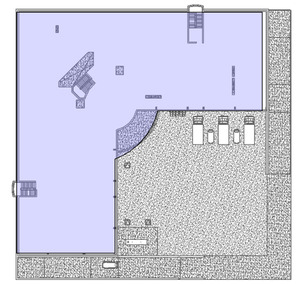
2 Duty Chillers × 55 TR = 110 TR

1 Standby Chiller × 55 TR

Total Chillers = 3 Nos. (2 Duty + 1 Standby)

SpaceName	Area	Height	Sensible Load	Total Load	CFMs	CFM_F	CFM_R	CFM_{EX}	System Name
	M^2	M	TR	TR					
00-63-Producer	42	2.8	5.3	8.6	1070	100%	0	819	00-01-AHU
00-57-Operation	44.4	2.8	7.5	12.2	1507	100%	0	1073	00-02-AHU
00-58-Operation	44.1	2.8	6.7	11	1361	100%	0	944	00-03-AHU
00-64-Recovery	52	2.8	4.7	7.7	943	100%	0	522	00-04-AHU
00-60-Sterilize	7	2.8	6	9.6	146.6	100%	0	71.5	00-05-AHU
00-61-Sterilize	6	2.8			113	100%	0	38	
00-59-Clean	56	2.8			900	100%	0	1650	
00-44-Isolation	11.5	2.8	1.5	2.5	221	100%	0	372.4	00-06-AHU
00-42-Antiroom	3.1	2.8			90	100%	0	240.1	
00-41-ICU	120.4	2.8	15	25	3064	100%	0	1747	00-07-AHU
00-48-Rest	8.4	2.8	0.4	0.65	110	10%	99	NR	00-08-FCU
00-49-Services	10.3	2.8			105	10%	95	NR	
00-50-Corridor	92.1	2.8	1.5	1.95	808	10%	727	NR	00-09-FCU
00-51-Dirty Aera	6.7	2.8	0.5	0.62	90	10%	80	NR	00-10-FCU
00-52-Equip Sterilize	7.9	2.8			135	10%	122	NR	
00-45-Corridor	94.4	2.8	2.2	2.5	1111	10%	1000	NR	00-11-FCU
00-26-Reception	104.9	2.8	2.3	3	1107	10%	996	NR	00-12-FCU
00-33-Corridor	48.4	2.8	0.76	1.1	394	10%	355	NR	00-13-FCU
00-28-Corridor	67.8	2.8	1.27	1.5	1092	10%	983	NR	00-14-FCU
00-38-Clean Unit	9.3	2.8	0.36	0.6	177	10%	159	NR	00-15-FCU
00-36-Medication	4.9	2.8	0.1	0.2	59	10%	53	NR	00-16-FCU
00-35-Assistants Room	12.6	2.8	0.45	0.6	244	10%	220	NR	00-17-FCU
00-32-Doctor Room	8.7	2.8	0.8	1	200	10%	180	NR	00-18-FCU
00-31-Staff Room	8.9	2.8			200	10%	180	NR	
00-27-Hospital Director	32.7	2.8	0.8	1	400	10%	360	NR	00-19-FCU
00-24-Meeting Room	28.1	2.8	0.76	1.2	394	10%	355	NR	00-20-FCU
00-20-Assistant Director	31.2	2.8	0.76	0.9	390	10%	351	NR	00-21-FCU
00-15-Medical Record	55.2	2.8	1.3	1.7	666	10%	599	NR	00-22-FCU
00-01-Accounting	30.2	2.8	0.8	1	390	10%	351	NR	00-23-FCU
00-03-DCFW	77.3	2.8	1.8	2.4	930	10%	837	NR	00-24-FCU
00-10-Rest	7.1	2.8	0.36	0.62	78	10%	70	NR	00-25-FCU
00-08-Rest	8.2	2.8			111	10%	100	NR	
00-11-Corridor	171.3	2.8	3.5	3.6	2760	10%	2484	NR	00-26-FCU
00-09-DCM	76.9	2.8	1.8	2.44	926	10%	833	NR	00-27-FCU
00-14-Head Nurse	11.3	2.8	0.36	0.5	100	10%	90	NR	00-28-FCU
00-13-Services	8.8	2.8			86	10%	77	NR	
00-25-Corridor	52.5	2.8	1	1.16	846	10%	761	NR	00-29-FCU
00-21-Day Care	26	2.8	0.8	1.3	400	10%	360	NR	00-30-FCU

Key Plan:



Legends:

CFMs	Cubic Feet per Minute Supply
CFM_F	Cubic Feet per Minute Fresh
CFM_{EX}	Cubic Feet per Minute Exhaust
AUH	Air Handling Unit
FCU	Fan Coil Unit

Notes:

Cooling Load Calculations are based on HAP software.

Supply Air Quantities are based on calculated room cooling loads.

MOHAMED SHABAN
MECHANICAL ENGINEER

Project: Hospital

Owner: ENGX

Date: 25/9/2026

Designer: M.SHABAN

Checked by: Taha Hussein

DRAWING: Ground Floor

DISCIPLINE: HVAC

Scale: 1:70